PREP 1, LESSONS 1–2

Can this data be trusted?

Mean and uncertainty; random and systematic errors; variables and choosing a graph

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Define the resolution of a measuring instrument. DN 852 [1]

The smallest change the instrument can detect.

2 Describe the effect of a random error on a set of readings. DN 860 [1]

It scatters the readings unpredictably (some too high, some too low).

3 State how the effect of random error can be reduced. DN 864 [1]

Take more repeats and calculate a mean.

Key ideas LESSONS 1–2

Show your working

4 A student times a toy car rolling down a ramp three times: **2.31 s**, **2.45 s** and **2.38 s**.

a Calculate the mean time. [2]

mean = total ÷ number of trials = (2.31 + 2.45 + 2.38) ÷ 3 = 7.14 ÷ 3 ✓

= 2.38 s ✓

b Calculate the uncertainty in the mean. Use uncertainty = (largest − smallest) ÷ 2 [1]

(2.45 − 2.31) ÷ 2 = 0.14 ÷ 2 = 0.07 s ✓

5 A balance reads 0.2 g when nothing is on it. The student uses it to measure the mass of a ball five times.

a This causes a **random** / **systematic** error. [1]

b Explain why taking more repeats will not fix this error. [1]

Every reading is shifted by the same amount (0.2 g too high), so the mean is shifted too ✓

6 A student investigates how the **type of surface** affects the **distance** a toy car travels.

a The dependent variable is the distance travelled (the type of surface / the distance travelled / the mass of the car) [1]

b Draw a ring around the correct graph: **bar chart** / **line graph** Give the reason. [1]

The independent variable (type of surface) is categoric ✓

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